

HVAC parts list - 8835 Semiahmoo Dr, Blaine WA 98230

SFR2026-00068 model **8835-semiahmoo-hvac** printed 2026-09-25 ducted_multizone heat 23,560 Btuh cooling 1.81 t 1 outdoor 0 heads 2 ducted units 20 registers line sets 111 ft duct 522 ft vent 511 ft 19 days on site
Design 19 F / 78 F (blaine: WSEC Table RC-1 Bellingham IAP 19 F / 78 F [S]; MJ8 Table 1A 23 F / 76 F [V] (Fraser outflow country - STAND-IN on the Bellingham row)). Foundation crawl, line sets concealed, ducts inside, ventilation erv. Quantities only - the prices are set on the price list in the app.

Loads

heating 23,560 Btuh (6.9 Btuh/sf) · cooling 20,857 + 920 latent · WSU sheet 39,604 Btuh · infiltration 0.6 ACH50 -> 0.054 ACH at design · ventilation 64.3 cfm (erv)

room	storey	sf	heat Btuh	cool Btuh	glass sf	ext wall ft
DEN	upper	452	2,828	4,132	139	38
ROOM	upper	491	2,741	2,375	93	60
BEDROOM	upper	199	1,633	597	11	51
LANDING	upper	64	1,498	2,485	91	18
LIVING ROOM	main	208	1,277	416	36	30
BEDROOM	upper	214	1,071	434	0	33
LAUNDRY	main	65	923	868	36	20
main/R24	main	7	919	548	14	14
PRIMARY	main	232	900	481	0	31
GREAT ROOM	main	113	883	282	0	30
ENTRY	main	93	857	929	34	18
GREAT ROOM	main	298	843	182	0	25
main/R13	main	18	824	1,457	54	9
main/R3	main	15	822	1,286	48	13
YOGA	main	121	812	746	39	12
main/R1	main	9	704	747	28	19
main/R7	main	17	618	129	0	12
main/R11	main	18	552	789	29	9
KITCHEN	main	171	513	1,438	0	12
DINING	main	107	380	270	0	6
CLOSET	main	55	239	14	0	8
HALL	main	106	216	73	0	0
ENSUITE	upper	21	151	21	0	7
BATH	upper	21	122	21	0	5
COATS	main	77	110	17	0	0
BATH	main	39	100	12	0	2
W.C.	main	25	99	23	0	2
HALL	upper	15	59	14	0	2
hall	main	16	35	13	0	0
POWDER	main	9	21	8	0	0
main/R25	main	8	19	7	0	0
VANITY	main	17	19	0	0	0
shower	main	14	18	1	0	0
HALL	main	14	15	0	0	0
BROOM	main	8	9	0	0	0
PANTRY	main	7	8	0	0	0

The system

unit house - ducted_multizone: heat 23,560 Btuh, capacity at design 55,992 Btuh
outdoor eq_mz_60 60k (multi) on the N wall, at the electrician's Heat pump condenser (LIVING ROOM, electrical/jobs/semiahmoo/devices.json); 111 ft of line set
indoor eq_duin_18 18k in LAUNDRY (main), 525 cfm
indoor eq_duin_12 12k in HALL (upper), 458 cfm

register room	storey	cfm	duct
YOGA	main	53	6 in
ENTRY	main	66	6 in
BATH	main	30	5 in

LIVING ROOM	main	83	6 in
GREAT ROOM	main	55	6 in
HALL	main	30	5 in
KITCHEN	main	51	6 in
KITCHEN	main	51	6 in
DINING	main	30	5 in
PRIMARY	main	59	6 in
GREAT ROOM	main	58	6 in
hall	main	30	5 in
DEN	upper	82	6 in
DEN	upper	82	6 in
ROOM	upper	49	5 in
ROOM	upper	49	5 in
LANDING	upper	49	5 in
LANDING	upper	49	5 in
BEDROOM	upper	39	5 in
BEDROOM	upper	59	6 in
return	storey	cfm	duct
ENTRY	main	596	12 in
HALL	upper	458	12 in
			grille
			16x20
			14x20

Runs (measured, before waste)

system / size	LF
dryer 4	4
erv_exhaust 4	39
erv_exhaust 5	81
erv_outdoor 6	22
erv_supply 4	191
erv_supply 6	111
exhaust 4	52
hood 7	9
refrigerant 0.25x0.375	39
refrigerant 0.25x0.5	72
return 12	17
supply 5	194
supply 6	205
trunk 12	106

Equipment

Equipment item	measured	order	unit	note
Ducted indoor unit (slim / multi-position) 12k	1	1	EA	ducted indoor unit 12k in HALL (upper; rule: first CLOS CLST HALL LANDING room)
Ducted indoor unit (slim / multi-position) 18k	1	1	EA	ducted indoor unit 18k in LAUNDRY (main; rule: first LAUNDRY UTIL room)
ERV / HRV - balanced whole-house, 50-150 cfm (SRE per the credit)	1	1	EA	unit house: VENTS Frigate ERV 150 - 64 cfm continuous (balanced, ducted to every habitable room: Csystem 1.0),
Multi-zone outdoor unit - 60k, cold-climate, up to 8 indoor units	1	1	EA	multi-zone outdoor unit 60k (~55,992 Btuh at 19 F), at the electrician's Heat pump condenser (LIVING ROOM, ele
Thermostat - programmable, heat pump with aux-heat lockout (WSEC R403.1.1 / R403.1.2)	2	2	EA	WSEC R403.1.1 programmable; heat-pump aux lockout R403.1.2

Rough-in

Condensate item	measured	order	unit	note
Condensate line 3/4 in PVC (or 5/8 vinyl with the line set)	25	28	LF	air handler condensate to an approved discharge
Condensate trap + cleanout tee	2	2	EA	trap + cleanout tee

Secondary drain pan + float switch (M1411.3.1: equipment above finished space)	1	1	EA	upper unit sits over finished space (M1411.3.1)
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Ductwork

item	measured	order	unit	note
Duct hanger strap / saddle (flex every 4 ft, rigid every 10 ft)	125	125	EA	trunk
Duct mastic (gal) + UL-181 foil tape	10	10	EA	mastic by the gallon per 100 ft of duct
Flex duct R-8 5 in	194.1	204	LF	BATH: 30 cfm · HALL: 30 cfm · DINING: 30 cfm · hall: 30 cfm · ROOM: 49 cfm · LANDING: 49 cfm
Flex duct R-8 6 in	204.9	216	LF	YOGA: 53 cfm · ENTRY: 66 cfm · LIVING ROOM: 83 cfm · GREAT ROOM: 55 cfm · KITCHEN: 51 cfm · PRIMARY: 59 cfm
Register boot (4x10 / 4x12 floor or 6x10 ceiling)	20	20	EA	
Return duct 12-16 in (panned joist not allowed for a new return)	16.8	18	LF	main central return 12 in, 596 cfm, 16x20 filter grille · upper central return 12 in, 458 cfm, 14x20 filter gr
Supply + return plenums (fabricated, sealed, lined where required)	2	2	EA	supply + return plenums at the unit
Supply trunk - galvanized rectangular / 10-14 in round (extended plenum)	106.4	112	LF	main: 12 in equivalent, 596 cfm - reducing past 24 ft · upper: 12 in equivalent, 458 cfm - reducing past 24 ft
Takeoff / starting collar with balancing damper	20	20	EA	
Transfer grille pair / jumper duct (closed-door pressure relief)	4	4	EA	PRIMARY: 59 cfm needs ~32 sq in free area at 3 Pa - transfer grille pair or jumper duct · DEN: 163 cfm needs ~

Refrigerant piping

item	measured	order	unit	note
Anti-vibration pads (set of 4)	1	1	EA	
Communication cable 14/4 stranded (outdoor unit to each indoor unit)	111.1	123	LF	with the line set
Equipment pad - composite, sized to the outdoor unit	1	1	EA	outdoor unit pad
Line set 1/4 x 1/2 insulated	71.9	80	LF	18k indoor unit, 72 ft
Line set 1/4 x 3/8 insulated (pre-flared or brazed)	39.2	44	LF	12k indoor unit, 39 ft
Outdoor unit stand / wall bracket (snow and drainage clearance)	1	1	EA	snow / drainage stand
Wall penetration - 3 in sleeve, sealed and flashed	1	1	EA	rim penetration for the line set

Support + protection

item	measured	order	unit	note
Fire caulk / sealant (tube)	1	1	EA	fire caulk / sealant at the penetrations

Ventilation and exhaust

Ventilation

item	measured	order	unit	note
Bath fan exhaust duct 4 in insulated flex (50 cfm fan: <= 25 ft, Table M1505.4.4.2)	52.5	58	LF	BATH: 50 cfm, 5 ft, 2 elbows -> 4 in flex (Table M1505.4.4.2 allows 25 ft) · W.C.: 50 cfm, 7 ft, 2 elbows -> 4
Dryer duct 4 in 90 elbow	2	2	EA	
Dryer duct 4 in rigid metal, smooth wall (M1502.4.2)	4.5	5	LF	from the electrician's dryer outlet in LAUNDRY: 4 ft
Dryer vent box (recessed)	1	1	EA	
Dryer wall cap with damper, no screen (M1502.3)	1	1	EA	
ERV duct 5-6 in (supply to bedrooms / living, exhaust from baths / kitchen / laundry)	422.9	466	LF	supply trunk, main · supply YOGA · supply LIVING ROOM · supply GREAT ROOM · supply DINING · supply PRIMARY
ERV intake / exhaust hood 6 in (intake >= 10 ft from any exhaust, R303.5 / M1505.2)	2	2	EA	outdoor-air intake - 12 ft apart on the S wall (M1504.3: exhaust >= 10 ft from an intake) · exhaust outlet - 1
ERV outdoor-air / exhaust duct, insulated with vapour jacket (to the hoods)	22.1	25	LF	ERV outdoor-air intake to its wall hood (insulated, vapour jacket) · ERV exhaust outlet to its wall hood (insu
ERV supply / exhaust grille or valve 4-6 in	15	15	EA	YOGA · LIVING ROOM · GREAT ROOM · DINING · PRIMARY · DEN
Range hood duct 6-10 in rigid, sealed (M1503.3)	9	10	LF	7 in rigid for a 400 cfm hood at the electrician's range outlet (electrical/jobs/semiahmoo/devices.json) (M150
Range hood wall / roof cap with damper	1	1	EA	7 in wall cap with damper
Wall cap 4 in with backdraft damper	6	6	EA	BATH: wall cap with backdraft damper · W.C.: wall cap with backdraft damper · shower: wall cap with backdraft

Trim

Registers + grilles

item	measured	order	unit	note
Filters MERV 8-13 (first set)	2	2	EA	first filter set
Floor register 4x10 / 4x12	20	20	EA	YOGA · ENTRY · BATH · LIVING ROOM · GREAT ROOM · HALL
Return / filter grille 20x25 (or 14x30)	2	2	EA	ENTRY (main): 16x20 at <= 300 fpm · HALL (upper): 14x20 at <= 300 fpm

Start-up, tests and design

Commissioning

item	measured	order	unit	note
Air balance of supply registers (per ducted system)	2	2	EA	register balance
Equipment labels, WSEC certificate data, owner hand-over	1	1	EA	per dwelling unit
Load calculation + equipment selection for the permit (Manual J / S or the WSU sizing form)	1	1	EA	room-by-room loads for the permit (this takeoff is the draft)
Start-up, charge verification, controls set-up (per outdoor unit)	1	1	EA	60k system
Ventilation flow test - whole-house + local exhaust (WA M1505.4.3 / M1505.4.4)	1	1	EA	unit house: whole-house 64 cfm + every local exhaust measured at the grille (WA M1505.4.3 / M1505.4.4)

Refrigerant piping

item	measured	order	unit	note
Nitrogen pressure test + evacuation to 500 micron (per system)	1	1	EA	per refrigerant system
Refrigerant - additional charge beyond the pre-charge (lb)	2.1	2.1	lb	33 oz past the pre-charge [U: the maker's oz / ft]

The checker

DATA plan set: credit 2.3 (heat-recovery ventilation) AND a whole-house exhaust fan are both on the set -> hrv (if the ERV/HRV is the whole-house system the fan is local exhaust only; if the fan is the whole-house system the credit fails - ask the designer)

PRACTICE unit house: the 60k outdoor unit carries 55,992 Btuh at 19 F against a 23,560 Btuh load - 2.4x AS SPECIFIED on the plan: ask how it was sized (Manual S 2023 wants >= 1.00 x at design and the minimum output <= 0.80 x; WA M1401.3 lets variable-capacity equipment run over, but not forever)

DATA design temperature: 19 F is a stand-in (WSEC Table RC-1 Bellingham IAP 19 F / 78 F [S]; MJ8 Table 1A 23 F / 76 F [V] (Fraser outflow country - STAND-IN on the Bellingham row)) - pull WSEC Table RC-1 for blaine before this goes out

PRACTICE electrical: the 60k multi outdoor unit typically needs a 50 A circuit [U: the nameplate MOCP governs]; the electrical job carries 40 A

note: tracer fragments merged into one space (same name, touching): hall (main: 2 pieces, 30 sf)

note: the electrician put the air handler in WORKBENCH / SHOP (outside the conditioned space) - not followed; tell the electrician where it moved

note: bath fans: 6 read from the electrical skill (electrical/jobs/semiahmoo/devices.json) - the electrician supplies and wires them; HVAC ducts and caps each one

note: unit house: ERV supplies 10 habitable rooms and exhausts 5 wet rooms on a trunk-and-branch layout - 423 ft of 4-6 in duct

note: equipment must meet HSPF2 >= 9.4 (the WSEC credit); design temperature 19 F <= 23 F: a NEEP-listed cold-climate variable-capacity unit (the 3.6 note)

note: the 60k nominal is over the WSU sizing sheet's heat-pump maximum (49,505 Btuh = 1.25 x its 39,604) - if the permit used the WSU sheet, the reviewer can ask

note: refrigerant: a split system installed from 1/1/2026 is an A2L (R-454B / R-32) unit under the EPA Technology Transitions rule; R-410A equipment is out. A2L units carry leak detection / mitigation per their listing - it rides with the equipment line

note: a gas-fuel fireplace is on the set - not in this bid (the fireplace installer's); a sealed direct-vent unit needs no combustion air from the house

Labour and inspections

crew of 2: set 18.0 h / 3 d, rough 79.3 h / 13 d, trim 3.2 h / 1 d, startup 8.0 h / 2 d - 19 days at 6.25 h. crew minutes are PLACEHOLDERS (no task-level install hours were found: references/research-hvac-market.md 3.3) - set them from a sub or the crew

Mechanical rough-in (cover) - after framing, before insulation; line sets pressure-tested and in place; ducts, boots, exhaust ducts and caps; firestops at the garage separation

Duct leakage test - rough-in or final; total leakage at 25 Pa per WSEC R403.3.5 - waived only when every duct and the air handler are inside the envelope

Ventilation flow test - before final; whole-house rate and every local exhaust measured (WA M1505.4)

Final mechanical - equipment set, started, labelled; equipment listing / HSPF2 on the certificate, controls, condensate, clearances

permit: City of Blaine residential mechanical fee schedule - NOT FOUND in the 9/25 research (references/research-hvac-code.md 8): price it from the county sheet